

Geometry & Coordinate Geometry



Distance and midpoint

- 1 Find the distance between $(1, 2)$ and $(4, 6)$.
 - 2 Find the midpoint of $(2, -3)$ and $(8, 5)$.
 - 3 Find the distance between $(-2, 1)$ and $(3, -11)$.
 - 4 Find the midpoint of $(-4, 7)$ and $(6, -3)$.
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Gradient (slope)

- 5 Find the gradient of the line through $(1, 2)$ and $(5, 10)$.
 - 6 Find the gradient of the line through $(-3, 4)$ and $(2, -6)$.
 - 7 Find the gradient of the line through $(0, 0)$ and $(6, 9)$.
 - 8 State whether the line through $(1, 1)$ and $(4, 1)$ is horizontal, vertical, or neither.
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Equations of lines

- 9 Find the equation of the line with gradient 2 passing through $(3, 5)$, in the form $y = mx + c$.
 - 10 Find the equation of the line through $(0, 4)$ and $(2, 10)$.
 - 11 Find the x - and y -intercepts of the line $2x + 3y = 12$.
 - 12 Write the equation of the line parallel to $y = 4x - 1$ passing through $(2, 9)$.
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Parallel and perpendicular lines

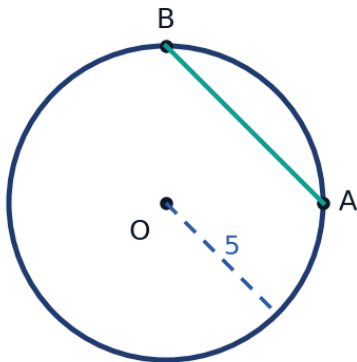
- 13 State the gradient of a line perpendicular to a line with gradient 5.
- 14 Determine whether the lines $y = 2x + 3$ and $y = 2x - 7$ are parallel, perpendicular, or neither.
- 15 Find the equation of the line perpendicular to $y = -\frac{1}{3}x + 2$, passing through $(3, 4)$.

The equation of a circle

- 16 Write the equation of a circle centered at the origin with radius 6.
- 17 Write the equation of a circle centered at $(2, -3)$ with radius 5.
- 18 Find the radius and center of the circle $(x - 1)^2 + (y + 4)^2 = 49$.
- 19 Find the equation of the circle centered at the origin that passes through $(3, 4)$.

Circles: chords and arcs

- 20 Points A and B lie on a circle of radius 5 centered at O , with central angle $\angle AOB = 90^\circ$. Find the length of chord AB , using $AB = 2r \sin\left(\frac{\theta}{2}\right)$.



- 21 A chord of a circle with radius 10 cm is 12 cm long. Find the distance from the center to the chord.
- 22 Find the circumference of a circle with radius 9 cm, in terms of π .

Polygon angle sums

- 23 Find the sum of the interior angles of a hexagon (6 sides).
- 24 Find the measure of each interior angle of a regular pentagon.
- 25 A polygon has an interior angle sum of 1440° . Find the number of sides.

Perimeter and area

- 26 A rectangle has length 12 and width 7. Find its perimeter and area.
- 27 A right triangle has legs 6 and 8. Find its area and hypotenuse.
- 28 A regular hexagon has side length 4, with area $A = \frac{3\sqrt{3}}{2}s^2$. Find its area in exact form.

Coordinate geometry: identifying shapes

- 29 Show that the triangle with vertices $A(0, 0)$, $B(4, 0)$, $C(4, 3)$ is a right triangle.
- 30 Determine whether the points $A(-2, 1)$, $B(2, 4)$, $C(5, 0)$ form a right triangle, using the squared side lengths.
- 31 Determine whether the triangle with vertices $A(1, 2)$, $B(5, 2)$, $C(3, 6)$ is isosceles.

The perpendicular bisector of a segment

- 32 Find the equation of the perpendicular bisector of the segment joining $(2, 3)$ and $(8, 7)$.
- 33 Find the equation of the perpendicular bisector of the segment joining $(0, 0)$ and $(6, 8)$.
- 34 A line segment has endpoints $(1, 5)$ and $(7, 5)$. Find the equation of its perpendicular bisector.

Word problems: geometry on a coordinate grid

- 35 This question has two parts. A city park is designed on a coordinate grid (in meters), with two gates at $A(0, 0)$ and $B(80, 60)$.
- a Find the straight-line distance between the two gates.
- b A fountain is placed at the midpoint between the two gates. Find its coordinates.

- 36 This question has two parts. A circular irrigation sprinkler is centered at $(5, 5)$ on a farm's coordinate grid (in meters) and has a radius of 12 meters.
- a Write the equation of the circle covered by the sprinkler.
 - b Determine whether a tree at $(14, 9)$ lies inside, on, or outside the watered region.
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MCQ — Distance and midpoint

- 37 Find the distance between $(2, 3)$ and $(6, 6)$.
- a. 5
 - b. 25
 - c. $\sqrt{7}$
 - d. 7
- 38 Find the midpoint of $(3, -4)$ and $(9, 6)$.
- a. $(6, 1)$
 - b. $(6, 2)$
 - c. $(12, 2)$
 - d. $(3, 10)$
- 39 Find the distance between $(-3, 2)$ and $(5, -13)$.
- a. 17
 - b. 15
 - c. 19
 - d. 289

40 Find the midpoint of $(-5, 8)$ and $(7, -2)$.

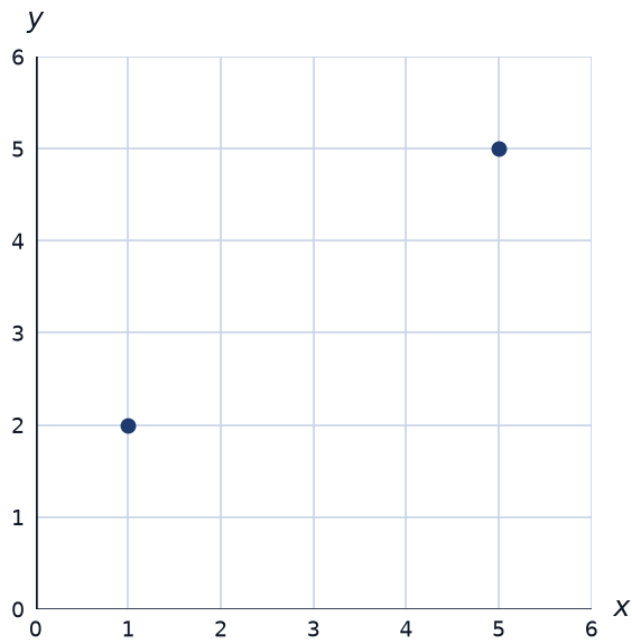
a. $(6, 1)$

b. $(1, 6)$

c. $(2, 6)$

d. $(1, 3)$

41 Find the distance between the two points shown.



a. $\sqrt{5}$

b. 4

c. 3

d. 5

- 42 Find the midpoint of the segment joining the two points shown.



- a. (4, 3)
- b. (2, 6)
- c. (2, 3)
- d. (8, 6)

MCQ — Gradient (slope)

- 43 Find the gradient of the line through (2, 3) and (6, 11).

- a. 8
- b. 0.5
- c. 4
- d. 2

- 44 Find the gradient of the line through $(-2, 5)$ and $(3, -10)$.

- a. $-\frac{1}{3}$
- b. -15
- c. -3
- d. 3

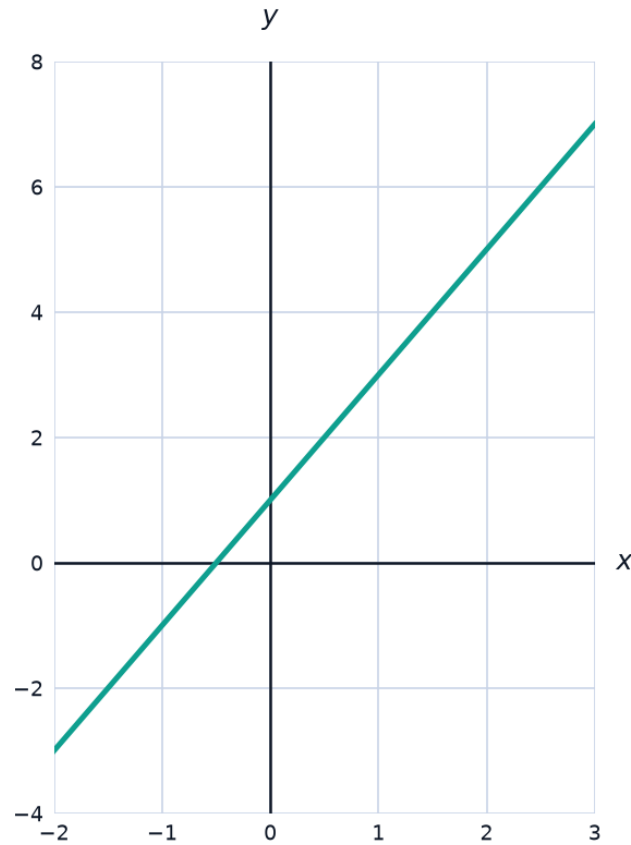
- 45 Find the gradient of the line through $(0, 0)$ and $(8, 12)$.
- a. 1.5
 - b. $\frac{8}{12}$
 - c. 96
 - d. $\frac{2}{3}$
- 46 State whether the line through $(2, 3)$ and $(2, 9)$ is horizontal, vertical, or neither.
- a. Cannot be determined
 - b. Neither
 - c. Horizontal
 - d. Vertical
-

MCQ — Equations of lines

- 47 Find the equation of the line with gradient 3 passing through $(2, 4)$, in the form $y = mx + c$.
- a. $y = 3x + 4$
 - b. $y = 3x - 2$
 - c. $y = 3x + 2$
 - d. $y = 2x - 3$
- 48 Find the equation of the line through $(0, 5)$ and $(3, 14)$.
- a. $y = 5x + 3$
 - b. $y = 9x + 5$
 - c. $y = 3x + 5$
 - d. $y = 3x - 5$

- 49 Find the x - and y -intercepts of the line $3x + 4y = 24$.
- a. $x = 8, y = 6$
 - b. $x = 6, y = 8$
 - c. $x = 24, y = 24$
 - d. $x = 4, y = 3$
- 50 Write the equation of the line parallel to $y = 5x - 2$ passing through $(1, 8)$.
- a. $y = 5x + 3$
 - b. $y = -\frac{1}{5}x + 3$
 - c. $y = 5x - 3$
 - d. $y = 5x + 8$
- 51 Find the equation of the line through $(1, 3)$ and $(4, 12)$.
- a. $y = 3x + 3$
 - b. $y = x + 3$
 - c. $y = 3x - 3$
 - d. $y = 3x$

- 52 The graph shows a line. Find its equation in the form $y = mx + c$ by reading the graph.



- a. $y = \frac{1}{2}x + 1$
- b. $y = 2x + 1$
- c. $y = x + 1$
- d. $y = 2x - 1$

MCQ — Parallel and perpendicular lines

- 53 State the gradient of a line perpendicular to a line with gradient 4.

- a. $-\frac{1}{4}$
- b. 4
- c. $\frac{1}{4}$
- d. -4

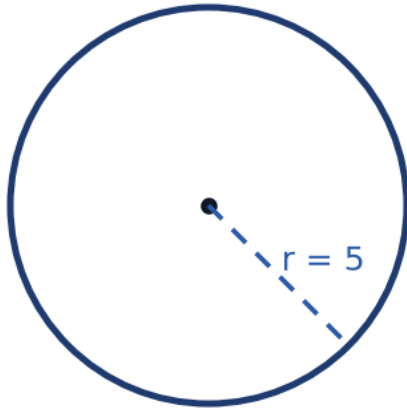
- 54 Determine whether the lines $y = 3x + 2$ and $y = 3x - 5$ are parallel, perpendicular, or neither.
- a. The same line
 - b. Neither
 - c. Perpendicular
 - d. Parallel
- 55 Find the equation of the line perpendicular to $y = -\frac{1}{4}x + 3$, passing through $(4, 5)$.
- a. $y = 4x - 5$
 - b. $y = 4x + 11$
 - c. $y = -\frac{1}{4}x + 6$
 - d. $y = 4x - 11$
- 56 Determine whether the lines $y = 2x + 1$ and $y = -0.5x + 4$ are parallel, perpendicular, or neither.
- a. The same line
 - b. Parallel
 - c. Neither
 - d. Perpendicular

MCQ — The equation of a circle

- 57 Write the equation of a circle centered at the origin with radius 8.
- a. $x^2 + y^2 = 64$
 - b. $x^2 + y^2 = 16$
 - c. $x^2 + y^2 = 8$
 - d. $(x - 8)^2 + (y - 8)^2 = 64$

- 58 Write the equation of a circle centered at $(3, -2)$ with radius 7.
- a. $(x - 3)^2 - (y + 2)^2 = 49$
 - b. $(x - 3)^2 + (y + 2)^2 = 49$
 - c. $(x - 3)^2 + (y + 2)^2 = 7$
 - d. $(x + 3)^2 + (y - 2)^2 = 49$
- 59 Find the radius and center of the circle $(x + 2)^2 + (y - 5)^2 = 81$.
- a. Center $(2, -5)$, radius 9
 - b. Center $(-2, 5)$, radius 9
 - c. Center $(-2, 5)$, radius 81
 - d. Center $(2, 5)$, radius 9
- 60 Find the equation of the circle centered at the origin that passes through $(6, 8)$.
- a. $x^2 + y^2 = 100$
 - b. $x^2 + y^2 = 10$
 - c. $x^2 + y^2 = 196$
 - d. $x^2 + y^2 = 48$

61 Write the equation of the circle shown, centered at the origin.



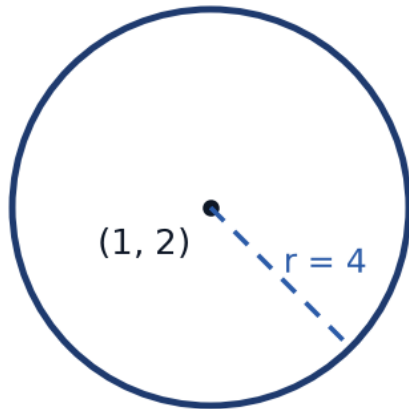
a. $x^2 + y^2 = 10$

b. $x^2 + y^2 = 5$

c. $x^2 + y^2 = 20$

d. $x^2 + y^2 = 25$

- 62 The circle shown is centered at $(1, 2)$ with the radius marked. Write its equation.



- a. $(x - 1)^2 + (y - 2)^2 = 8$
- b. $(x + 1)^2 + (y + 2)^2 = 16$
- c. $(x - 1)^2 + (y - 2)^2 = 16$
- d. $(x - 1)^2 + (y - 2)^2 = 4$

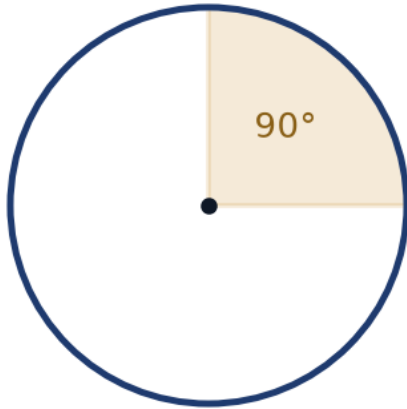
MCQ — Circles: chords and arcs

- 63 Points A and B lie on a circle of radius 8 centered at O , with central angle $\angle AOB = 60^\circ$. Find the arc length AB , in terms of π .

- a. $\frac{16\pi}{3}$
- b. $\frac{4\pi}{3}$
- c. 16π
- d. $\frac{8\pi}{3}$

- 64 A chord of a circle with radius 13 cm is 24 cm long. Find the distance from the center to the chord.
- a. 7
 - b. 1
 - c. 12
 - d. 5
- 65 Find the circumference of a circle with radius 11 cm, in terms of π .
- a. 121π
 - b. 44π
 - c. 22π
 - d. 11π

- 66 Find the arc length of the 90° sector shown, in terms of π .



- a. 12π
- b. 3π
- c. 1.5π
- d. 6π

MCQ — Polygon angle sums

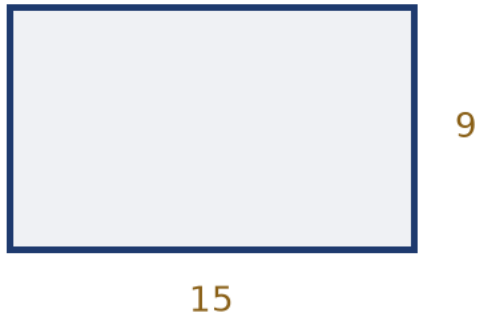
- 67 Find the sum of the interior angles of an octagon (8 sides).

- a. 720°
- b. $1,440^\circ$
- c. 900°
- d. $1,080^\circ$

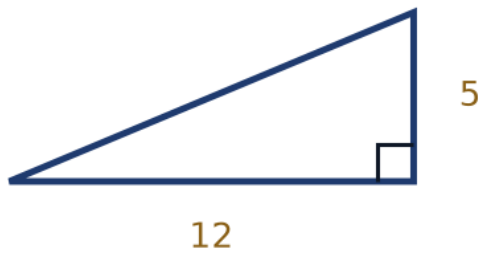
- 68 Find the measure of each interior angle of a regular hexagon.
- a. 140°
 - b. 135°
 - c. 108°
 - d. 120°
- 69 A polygon has an interior angle sum of $1,620^\circ$. Find the number of sides.
- a. 10
 - b. 12
 - c. 9
 - d. 11
- 70 Find the sum of the interior angles of a decagon (10 sides).
- a. $1,800^\circ$
 - b. $1,260^\circ$
 - c. $1,080^\circ$
 - d. $1,440^\circ$

MCQ — Perimeter and area

- 71 A rectangle has length 15 and width 9, as shown. Find its perimeter and area.



- a. $P = 135, A = 48$
 - b. $P = 48, A = 135$
 - c. $P = 24, A = 135$
 - d. $P = 48, A = 24$
- 72 A right triangle has legs 5 and 12, as shown. Find its area and hypotenuse.



- a. $A = 60, \text{hyp} = 13$
- b. $A = 30, \text{hyp} = 17$
- c. $A = 30, \text{hyp} = 13$
- d. $A = 30, \text{hyp} = 12$

- 73 A regular hexagon has side length 6, with area $A = \frac{3\sqrt{3}}{2}s^2$. Find its area in exact form.
- a. $36\sqrt{3}$
 - b. $108\sqrt{3}$
 - c. $54\sqrt{3}$
 - d. $18\sqrt{3}$
- 74 A regular pentagon has side length 8, with area $A \approx 1.72s^2$. Find its area to the nearest whole number.
- a. 110
 - b. 124
 - c. 96
 - d. 14

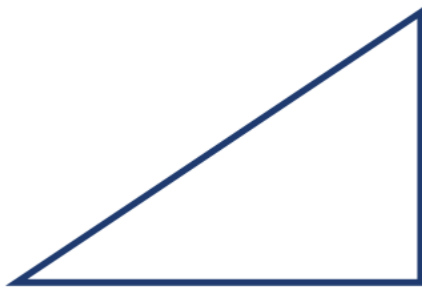
MCQ — Coordinate geometry: identifying shapes

- 75 Is the triangle with vertices $A(0, 0)$, $B(6, 0)$, $C(6, 4)$ a right triangle?
- a. No
 - b. Cannot be determined
 - c. Yes
 - d. Only if it is also isosceles
- 76 Determine whether the points $A(-3, 2)$, $B(1, 5)$, $C(4, 1)$ form a right triangle, using squared distances.
- a. Only if it is also equilateral
 - b. No
 - c. Cannot be determined
 - d. Yes

77 Determine whether the triangle with vertices $A(2, 3)$, $B(6, 3)$, $C(4, 7)$ is isosceles.

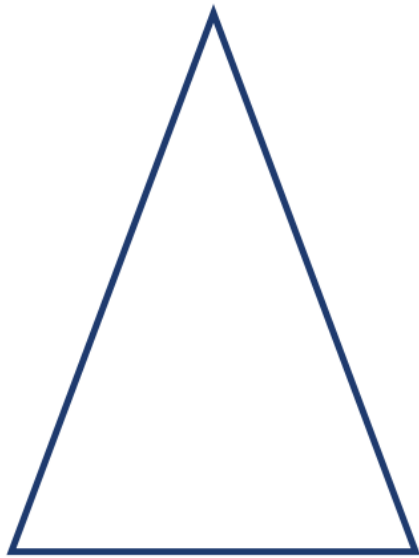
- a. No
- b. Only if it is also a right triangle
- c. Yes
- d. Cannot be determined

78 Is the triangle shown a right triangle?



- a. Only if it is also isosceles
- b. Cannot be determined
- c. No
- d. Yes

79 Is the triangle shown isosceles, scalene, or equilateral?



- a. Isosceles
- b. Cannot be determined
- c. Equilateral
- d. Scalene

MCQ — The perpendicular bisector of a segment

80 Find the equation of the perpendicular bisector of the segment joining (1, 4) and (7, 10).

- a. $y = -x + 11$
- b. $y = x - 11$
- c. $y = x + 11$
- d. $y = -x + 7$

- 81 A line segment has endpoints (2, 0) and (2, 8). Find the equation of its perpendicular bisector.
- a. $x = 4$
 - b. $y = 4$
 - c. $y = 2$
 - d. $x = 2$
- 82 A line segment has endpoints (3, 6) and (9, 6). Find the equation of its perpendicular bisector.
- a. $x = 3$
 - b. $y = 6$
 - c. $y = 9$
 - d. $x = 6$
- 83 Find the equation of the perpendicular bisector of the segment joining $(-2, 1)$ and $(4, 9)$.
- a. $y = -\frac{3}{4}x + \frac{23}{4}$
 - b. $y = -\frac{3}{4}x + 5$
 - c. $y = -\frac{4}{3}x + \frac{23}{4}$
 - d. $y = \frac{3}{4}x + \frac{23}{4}$
-

MCQ — Word problems: geometry on a coordinate grid

- 84 A city park is designed on a coordinate grid (in meters). Two gates are located at (10, 20) and (50, 60). Find the distance between the gates, to the nearest meter.
- a. 113
 - b. 40
 - c. 57
 - d. 80

- 85 A circular irrigation sprinkler is centered at $(8, 8)$ on a farm's coordinate grid (in meters) and has a radius of 15 meters. Write the equation of the circle it waters.
- a. $(x + 8)^2 + (y + 8)^2 = 225$
 - b. $(x - 8)^2 + (y - 8)^2 = 15$
 - c. $(x - 8)^2 + (y - 8)^2 = 225$
 - d. $(x - 8)^2 + (y - 8)^2 = 30$
- 86 A surveyor marks two points on a coordinate map (in km): a base camp at $(3, 4)$ and a summit at $(15, 4)$. Find the midpoint, to be used as a resupply station.
- a. $(6, 4)$
 - b. $(9, 8)$
 - c. $(9, 4)$
 - d. $(18, 8)$
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Multiple choice — easy

- 87 Find the distance between $(3, 2)$ and $(-2, 14)$.
- a. 15
 - b. 13
 - c. 17
 - d. 11
- 88 Find the distance between $(-7, -8)$ and $(2, 4)$.
- a. 15
 - b. 17
 - c. 21
 - d. 13

89 Find the distance between $(-5, 7)$ and $(-17, 23)$.

a. 22

b. 28

c. 20

d. 18

90 Find the distance between $(7, 4)$ and $(4, 0)$.

a. 8

b. 3

c. 7

d. 5

91 Find the distance between $(4, 0)$ and $(7, -4)$.

a. 5

b. 3

c. 7

d. 8

92 Find the midpoint of $(7, 6)$ and $(-9, 2)$.

a. $(0, 4)$

b. $(0, 5)$

c. $(-1, 5)$

d. $(-1, 4)$

93 Find the midpoint of $(8, 3)$ and $(0, -1)$.

a. $(4, 2)$

b. $(4, 1)$

c. $(5, 2)$

d. $(5, 1)$

94 Find the midpoint of $(-1, 0)$ and $(-1, 2)$.

a. $(0, 2)$

b. $(-1, 1)$

c. $(0, 1)$

d. $(-1, 2)$

95 Find the midpoint of $(-6, 7)$ and $(-6, 1)$.

a. $(-5, 5)$

b. $(-5, 4)$

c. $(-6, 5)$

d. $(-6, 4)$

96 Find the midpoint of $(-5, 2)$ and $(9, -4)$.

a. $(2, -1)$

b. $(3, 0)$

c. $(3, -1)$

d. $(2, 0)$

97 Find the gradient of the line through $(-3, 2)$ and $(-6, 3)$.

a. -3

b. $-\frac{1}{3}$

c. $\frac{1}{3}$

d. -1

98 Find the gradient of the line through $(-5, 2)$ and $(-3, 0)$.

a. 0

b. $-\frac{1}{2}$

c. -1

d. 1

99 Find the gradient of the line through $(-5, -5)$ and $(6, 17)$.

a. $\frac{1}{2}$

b. $\frac{24}{11}$

c. 2

d. -2

100 Find the gradient of the line through $(6, 1)$ and $(5, 2)$.

a. -3

b. -2

c. -1

d. 1

101 Find the gradient of the line through $(0, 1)$ and $(5, 21)$.

a. -4

b. $\frac{1}{4}$

c. 4

d. $\frac{22}{5}$

Multiple choice — medium

102 Find the equation of the line with gradient 4 passing through $(-4, 3)$, in the form $y = mx + c$.

a. $y = 4x + 21$

b. $y = 4x + 19$

c. $y = -4x + 19$

d. $y = 4x - 19$

103 Find the equation of the line with gradient -4 passing through $(-5, -5)$, in the form $y = mx + c$.

a. $y = -4x + 25$

b. $y = -4x - 23$

c. $y = 4x - 25$

d. $y = -4x - 25$

104 Find the equation of the line with gradient -3 passing through $(6, 2)$, in the form $y = mx + c$.

a. $y = -3x + 20$

b. $y = -3x + 22$

c. $y = 3x + 20$

d. $y = -3x - 20$

105 Find the equation of the line with gradient -3 passing through $(-4, 2)$, in the form $y = mx + c$.

a. $y = -3x - 10$

b. $y = -3x + 10$

c. $y = 3x - 10$

d. $y = -3x - 8$

106 Find the equation of the line with gradient 2 passing through $(-2, -3)$, in the form $y = mx + c$.

a. $y = 2x + 3$

b. $y = 2x + 1$

c. $y = -2x + 1$

d. $y = 2x - 1$

107 State the gradient of a line perpendicular to a line with gradient 4 .

a. $\frac{1}{4}$

b. $-\frac{1}{4}$

c. 4

d. $-\frac{1}{6}$

108 State the gradient of a line perpendicular to a line with gradient $\frac{4}{3}$.

a. $-\frac{1}{2}$

b. $\frac{4}{3}$

c. $-\frac{3}{4}$

d. $\frac{3}{4}$

109 State the gradient of a line perpendicular to a line with gradient $\frac{3}{5}$.

a. $-\frac{5}{3}$

b. -1

c. $\frac{3}{5}$

d. $\frac{5}{3}$

110 State the gradient of a line perpendicular to a line with gradient $\frac{2}{5}$.

a. $\frac{2}{5}$

b. $\frac{5}{2}$

c. $-\frac{5}{4}$

d. $-\frac{5}{2}$

111 State the gradient of a line perpendicular to a line with gradient $\frac{4}{7}$.

a. $\frac{7}{4}$

b. $-\frac{7}{6}$

c. $-\frac{7}{4}$

d. $\frac{4}{7}$

112 Write the equation of a circle centered at $(-1, -2)$ with radius 4.

a. $(x - 1)^2 + (y + 2)^2 = 16$

b. $(x + 1)^2 + (y - 2)^2 = 16$

c. $(x + 1)^2 + (y + 2)^2 = 16$

d. $(x + 1)^2 + (y + 2)^2 = 25$

113 Write the equation of a circle centered at $(-6, -2)$ with radius 3.

a. $(x + 6)^2 + (y + 2)^2 = 16$

b. $(x - 6)^2 + (y + 2)^2 = 9$

c. $(x + 6)^2 + (y - 2)^2 = 9$

d. $(x + 6)^2 + (y + 2)^2 = 9$

114 Write the equation of a circle centered at $(-4, 5)$ with radius 3.

a. $(x + 4)^2 + (y - 5)^2 = 9$

b. $(x - 4)^2 + (y - 5)^2 = 9$

c. $(x + 4)^2 + (y - 5)^2 = 16$

d. $(x + 4)^2 + (y + 5)^2 = 9$

115 Write the equation of a circle centered at $(1, 1)$ with radius 7.

a. $(x - 1)^2 + (y + 1)^2 = 49$

b. $(x - 1)^2 + (y - 1)^2 = 64$

c. $(x + 1)^2 + (y - 1)^2 = 49$

d. $(x - 1)^2 + (y - 1)^2 = 49$

116 Write the equation of a circle centered at the origin with radius 3.

a. $x^2 + (y - 1)^2 = 9$

b. $x^2 + y^2 = 16$

c. $x^2 + y^2 = 9$

d. $(x - 1)^2 + y^2 = 9$

Multiple choice — hard

117 Find the sum of the interior angles of a nonagon (9 sides).

- a. 1440°
- b. 1620°
- c. 1260°
- d. 1080°

118 Find the measure of each interior angle of a regular dodecagon.

- a. 30°
- b. 155°
- c. 150°
- d. 145°

119 Find the measure of each interior angle of a regular nonagon.

- a. 145°
- b. 140°
- c. 135°
- d. 40°

120 Find the sum of the interior angles of an octagon (8 sides).

- a. 1440°
- b. 1260°
- c. 1080°
- d. 900°

121 A polygon has an interior angle sum of 1080° . Find the number of sides.

- a. 10
- b. 9
- c. 7
- d. 8

122 A right triangle has legs 7 and 24. Find its area and hypotenuse.

- a. Area: 84. Hypotenuse: 26
- b. Area: 168. Hypotenuse: 25
- c. Area: 91. Hypotenuse: 25
- d. Area: 84. Hypotenuse: 25

123 A rectangle has length 14 and width 6. Find its perimeter and area.

- a. Perimeter: 20. Area: 84
- b. Perimeter: 40. Area: 98
- c. Perimeter: 42. Area: 84
- d. Perimeter: 40. Area: 84

124 A right triangle has legs 12 and 16. Find its area and hypotenuse.

- a. Area: 96. Hypotenuse: 21
- b. Area: 96. Hypotenuse: 20
- c. Area: 108. Hypotenuse: 20
- d. Area: 192. Hypotenuse: 20

125 A rectangle has length 7 and width 6. Find its perimeter and area.

- a. Perimeter: 28. Area: 42
- b. Perimeter: 26. Area: 49
- c. Perimeter: 26. Area: 42
- d. Perimeter: 13. Area: 42

126 A right triangle has legs 5 and 12. Find its area and hypotenuse.

- a. Area: 35. Hypotenuse: 13
- b. Area: 60. Hypotenuse: 13
- c. Area: 30. Hypotenuse: 13
- d. Area: 30. Hypotenuse: 14

127 Find the equation of the perpendicular bisector of the segment joining $(-1, 4)$ and $(5, 12)$.

- a. $y - 2 = -\frac{3}{4}(x - 8)$
- b. $y - 8 = \frac{3}{4}(x - 2)$
- c. $y - 8 = \frac{4}{3}(x - 2)$
- d. $y - 8 = -\frac{3}{4}(x - 2)$

128 Find the equation of the perpendicular bisector of the segment joining $(5, 4)$ and $(-1, 12)$.

- a. $y - 8 = \frac{3}{4}(x - 2)$
- b. $y - 8 = -\frac{4}{3}(x - 2)$
- c. $y - 8 = -\frac{3}{4}(x - 2)$
- d. $y - 2 = \frac{3}{4}(x - 8)$

129 Find the equation of the perpendicular bisector of the segment joining (4, 5) and (−2, 13).

a. $y - 1 = \frac{3}{4}(x - 9)$

b. $y - 9 = \frac{3}{4}(x - 1)$

c. $y - 9 = -\frac{3}{4}(x - 1)$

d. $y - 9 = -\frac{4}{3}(x - 1)$

130 Find the equation of the perpendicular bisector of the segment joining (2, 0) and (14, −16).

a. $y - 8 = \frac{3}{4}(x - (-8))$

b. $y - (-8) = -\frac{4}{3}(x - 8)$

c. $y - (-8) = -\frac{3}{4}(x - 8)$

d. $y - (-8) = \frac{3}{4}(x - 8)$

131 Find the equation of the perpendicular bisector of the segment joining (2, −2) and (−10, −18).

a. $y - (-10) = -\frac{3}{4}(x - (-4))$

b. $y - (-4) = -\frac{3}{4}(x - (-10))$

c. $y - (-10) = \frac{3}{4}(x - (-4))$

d. $y - (-10) = \frac{4}{3}(x - (-4))$